

Tight Frames, Efficient Packings — Some Feature Math*

Kaarel Hänni Jake Mendel

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1 Introduction

Throughout this piece, we will be assuming that we have a tuple of vectors $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$ — in words, a configuration of m vectors in $\mathbb{R}^n$, typically with $m > n$.¹ (We will ever-so-slightly abuse notation by letting $W \in (\mathbb{R}^n)^m \cong \mathbb{R}^{m \times n}$ also denote the $m \times n$ matrix which has W_i as its i th row.²) We would like to be able to talk about how “crammed each W_i feels” — that is, how much of its own dimension it has. To capture this, [Elh+22] proposed the following definition.

Definition 1.1. Given a vector configuration $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$, we define the *dimensionality* of the i th vector W_i as

$$\dim_i W = \frac{(W_i \cdot W_i)^2}{\sum_{j=1}^m (W_i \cdot W_j)^2}.$$

[Elh+22] demonstrated the usefulness of feature dimensionalities for empirically studying the feature configurations that show up in their toy model setup, but they did not provide much conceptual justification for the definition. One aim of this post is to make some more sense of this definition, mostly by proving it has some reasonable properties. A significant amount of sense-making was already done in [Zha22] and [Sch+23], to both of which this piece owes significantly in the questions we are asking.

Before we get to feature dimensionalities in Section 3, in Section 2, we discuss *tight frames* — vector configurations that are isotropic³ in a particular sense — which, in addition to being natural in their own right and having independent connections to feature geometry, have connections to feature dimensionalities.

Definition 1.2. We say a vector configuration $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$ is a *tight frame* if there is a $C_W \in \mathbb{R}$ such that for every $u \in \mathbb{R}^n$, we have

$$\sum_{i=1}^m \|\text{proj}_u W_i\|^2 = C_W.$$

More broadly, this piece is motivated by [Bec23] — in particular, we hope to contribute to the development of a mathematical theory of features in models. A lot of seeing further is about having the right concepts, and this post is in large part about examining/explaining/disseminating a few concepts that we think have the potential to be right.

*possible alternative subtitles: Bulletin of the European Auto Industry 2023; The Booming 60s in Soviet Architecture; Trends in Bodybuilding

¹For now, we will not require that these are unit vectors. But we are assuming they are all non-zero.

²This is essentially not an abuse of notation, because for this matrix W , we have $W_{ij} = (W_i)_j$. Note though that when we treat vectors as matrices, we treat them as column vectors by default. So in particular, even though the entries of W_i are also the entries of row i of W , when we treat W_i as a matrix, we still take it to be a column vector, not a row vector.

³<https://en.wikipedia.org/wiki/Isotropy>

2 A few frames on tight frames

Before we get to saying things about feature dimensionalities, let us introduce a notion of isotropy⁴ that, in addition to being natural in its own right — see Shayne Waldron’s 600-page monograph [Wal18], on which the half of the present section written after realizing there is a large literature on the mathematical object at hand is largely drawing from, for more on how it shows up in mathematics — turns out to be relevant to understanding feature dimensionality.

Definition 2.1. We say a vector configuration $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$ is a *tight frame* if there is a $C_W \in \mathbb{R}$ such that for every $u \in \mathbb{R}^n$, we have

$$\sum_{i=1}^m \|\text{proj}_u W_i\|^2 = C_W.$$

In fact, for a tight frame, we have

$$\sum_{i=1}^m \|W_i\|^2 = \sum_{i=1}^m \sum_{j=1}^n \|\text{proj}_{e_j} W_i\|^2 = \sum_{j=1}^n \sum_{i=1}^m \|\text{proj}_{e_j} W_i\|^2 = nC_W \implies C_W = \frac{1}{n} \sum_{i=1}^m \|W_i\|^2.$$

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Definition 2.2. For a tight frame, we call $C_W = \frac{1}{n} \sum_{i=1}^m \|W_i\|^2$ the *frame bound* of W .

Here’s the reason for calling C_W the frame ‘bound’ of W , as well as some background of independent interest⁶. Tight frames are a special case of frames — vector configurations W for which there are constants A_W and B_W , respectively called the *lower* and *upper frame bound*, such that $A_W \leq \sum_{i=1}^m \|\text{proj}_u W_i\|^2 \leq B_W$. This is sort of trivial for finite-dimensional spaces — any spanning set is a frame — but (1) it is interesting for infinite-dimensional spaces and (2) being a frame with some particular A_W and B_W (for instance, $A_W = 0.9$ and $B_W = 1.1$ ⁷) is non-trivial and useful. And well, a frame is said to be tight if both inequalities are tight — if its frame bounds A_W and B_W match.

In the rest of this section, we will provide characterizations of tight frames as configurations of vectors (Section 2.1), including ones that minimize a certain natural TMS-like loss function (Section 2.4), and as ‘overcomplete orthonormal bases’ (Section 2.2). We will also provide a way to construct tight frames (Section 2.3).

2.1 What does it mean for a vector configuration to be a tight frame?

Here are a few equivalent characterizations:

Lemma 2.1 (Equivalent definitions of tight frames). *The following are equivalent:*

1. $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$ is a tight frame with frame bound C_W .
2. For any $u, v \in \mathbb{R}^n$, we have

$$\sum_{i=1}^m \|\text{proj}_u W_i\|^2 = \sum_{i=1}^m \|\text{proj}_v W_i\|^2.$$

⁴<https://en.wikipedia.org/wiki/Isotropy>

⁵Alternatively, one can see this because for any configuration, $\mathbb{E}_{u \sim U(S^{n-1})} [\|\text{proj}_u W_i\|^2] = \frac{1}{n} \|W_i\|^2$, which one can see by picking any orthogonal basis $v_1, v_2, v_3, \dots, v_n$ with $W_i = v_1$ and $\|v_j\| = \|W_i\|$ for all $2 \leq j \leq n$, noting that $\sum_{j=1}^n \|\text{proj}_{v_j} W_i\|^2 = \|W_i\|^2$, and then using linearity of expectation and symmetry to conclude that $\|W_i\|^2 = \mathbb{E}_{u \sim U(S^{n-1})} [\sum_{j=1}^n \|\text{proj}_{v_j} W_i\|^2] = n \mathbb{E}_{u \sim U(S^{n-1})} [\|\text{proj}_u W_i\|^2]$. This additionally lets one describe a tight frame as a configuration with minimal maximal L^2 norm in a direction among all configurations with a certain total L^2 norm, and also a configuration with maximal minimal L^2 norm in a direction among all configurations with a certain total L^2 norm.

⁶and of contentment for those of us that like compositional definitions

⁷With W taken as a matrix with W_i as the rows, one can think of this as the requirement that the columns of W are nearly orthonormal — see [Tao07] if you want to go down a rabbit hole in this direction.

3. For any subspace $T \subseteq \mathbb{R}^n$ of dimension k ,

$$\sum_{i=1}^m \|\text{proj}_T W_i\|^2 = \frac{k}{n} \sum_{i=1}^m \|W_i\|^2.$$

4. For any unit vector $u \in \mathbb{R}^n$, the matrix W satisfies $\|Wu\|^2 = C_W$.

5. W has n equal nonzero singular values.

6. There is a set of orthonormal vectors $u_1, \dots, u_n \in \mathbb{R}^m$ and an orthonormal basis $v_1, \dots, v_n \in \mathbb{R}^n$ such that $W = \sqrt{C_W} \sum_{j=1}^n u_j v_j^T$.

7. $W^T W = C_W I$.

8.

We can also give a characterization in terms of properties of the data matrix:
as a linear operator, it is a rescaling of an isometry

If the set of vectors has mean 0, then being a tight frame is also equivalent to the data set being whitened (https://en.wikipedia.org/wiki/Whitening_transformation) up to rescaling — i.e., to each coordinate having the same sample variance and the covariance between any two different coordinates being 0.

2.2 A tight frame as a generalized orthonormal basis

Note that if W is an orthonormal basis, then $WW^T = I$, which⁸ implies $W^T W = I$. Thus, an orthonormal basis is a tight frame. Slightly more generally, any basis that consists of orthogonal vectors of the same norm is a tight frame. In at least this trivial sense, a tight frame is a generalization of an orthonormal basis. In fact, a tight frame is particularly a generalization of an orthonormal basis, and not a generalization of a basis — reversing the chain of implications from the start of this paragraph shows that any basis which is a tight frame must be a rescaling of an orthonormal basis (namely, rescaled by $\sqrt{C_W}$).

There is more than the above to a tight frame being a generalization of an orthonormal basis:

Proposition 2.1.1 (Tight frames as generalized orthonormal bases). *The following are equivalent:*

1. $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$ is a tight frame with frame bound $C_W \neq 0$.

2. (Pythagoras's identity) For every $u \in \mathbb{R}^n$,

$$\|u\|^2 = \frac{1}{C_W} \sum_{i=1}^m (u \cdot W_i)^2.$$

3. (Plancherel's identity) For any $u, v \in \mathbb{R}^n$,

$$u \cdot v = \frac{1}{C_W} \sum_{i=1}^m (u \cdot W_i) \cdot (v \cdot W_i)$$

4. (Parseval's identity) For any $u \in \mathbb{R}^n$,

$$u = \frac{1}{C_W} \sum_{i=1}^m (u \cdot W_i) W_i$$

Proof.

$$1 \implies 4. \quad \frac{1}{C_W} \sum_{i=1}^m (u \cdot W_i) W_i = \frac{1}{C_W} W^T (Wu) = \frac{1}{C_W} (W^T W)u = \frac{1}{C_W} C_W I u = u.$$

$$4 \implies 3. \quad u \cdot v = \left(\frac{1}{C_W} \sum_{j=1}^m (u \cdot W_j) W_j \right) \cdot v = \frac{1}{C_W} \sum_{i=1}^m (u \cdot W_i) \cdot (v \cdot W_i).$$

$$3 \implies 2. \quad \|u\|^2 = u \cdot u = \frac{1}{C_W} \sum_{i=1}^m (u \cdot W_i) \cdot (u \cdot W_i) = \frac{1}{C_W} \sum_{i=1}^m (u \cdot W_i)^2.$$

$$2 \implies 1. \quad \sum_{i=1}^m \|\text{proj}_u W_i\|^2 = \sum_{i=1}^m \frac{1}{\|u\|^2} (u \cdot W_i)^2 = \frac{C_W \|u\|^2}{\|u\|^2} = C_W.$$

□

⁸since for an orthonormal basis W , the matrix W is square

2.3 A way to construct tight frames

In this section, we will provide a way to construct a tight frame W with a given tuple of lengths, i.e., with $\|W_i\|^2 = a_i^2$ for some given constants $a_1, \dots, a_m \geq 0$ (subject to a necessary mild balancedness constraint on $a_1, \dots, a_m$). Other than this giving a way to generate examples, the existence of examples with particular length profiles will turn out to be relevant for understanding the extrema of certain optimization problems (Section 2.4 and Section 3.3). The method we provide here is essentially the one given by Lemma 2 in Appendix H of [Sch+23]. See [Cah+11] (Theorem 7) for a method for constructing all tight frames.

Theorem 2.2. *[There is a tight frame $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$ with $\|W_i\| = a_i$] $\iff$ [for all $i \in [m]$, we have $a_i^2 \leq \frac{1}{n} \sum_{j=1}^m a_j^2$].*

Proof of $\implies$. For a tight frame $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$ with frame bound C_W , we have⁹

$$\frac{1}{n} \sum_{j=1}^m a_j^2 = \frac{1}{n} \sum_{j=1}^m \|W_j\|^2 = C_W = \sum_{j=1}^m \|\text{proj}_{W_i} W_j\|^2 \geq \|W_i\|^2 = a_i^2.$$

□

Proof of $\impliedby$. It suffices to prove the existence of a tight frame W' with $\|W_i\| = a'_i = \frac{a_i}{\sqrt{\frac{1}{n} \sum_{j=1}^m a_j^2}}$, because rescaling back (i.e., $W_i = \sqrt{\frac{1}{n} \sum_{j=1}^m a_j^2} \cdot W'_i$) then provides a construction for the original $a_1, \dots, a_m$. Note that $a'_i = \frac{a_i}{\sqrt{\frac{1}{n} \sum_{j=1}^m a_j^2}} \leq 1$ and $\sum_{i=1}^m a_i'^2 = \frac{\sum_{i=1}^m a_i^2}{\frac{1}{n} \sum_{j=1}^m a_j^2} = n$. So, forgetting the primes, it suffices to prove the existence of an $m \times n$ matrix W whose n columns are orthonormal and whose m rows have norms $a_1, \dots, a_m$ which are guaranteed to be at most 1 and satisfy $\sum_{i=1}^m a_i^2 = n$.

Note that $\sum_{i=1}^m a_i^2 = n$ and $a_i \leq 1$ together imply that $m \geq n$. We prove the existence of a W with orthonormal columns and any given such squared row norms $a_1, \dots, a_m$ by induction on $n + m$. The base case $(n, m) = (0, 0)$ is trivial. For the inductive step, with $m \geq n$ and $m > 0$, we split into two cases:

1. If $2n > m$, then $m - n < n$, so by our inductive hypothesis, there is a matrix W' for $(n', m) = (m - n, n)$ with $a_1'^2 = 1 - a_1^2, \dots, a_m'^2 = 1 - a_m^2$ (since $0 \leq a'_i \leq 1$ and $\sum_{i=1}^m a_i'^2 = m - n$). Since the columns of W' are orthonormal, they can be completed into a basis — i.e., we can add n columns to W' to create an orthogonal matrix $U = [W'|W]$ ¹⁰ Note that W is then a matrix with orthonormal columns and with row norms $a_1, \dots, a_m$, as desired.¹¹
2. If $2n \leq m$, then there are $i \neq j$ with $a_i^2 + a_j^2 \leq 1$.¹² Then, by the inductive hypothesis, there is a matrix W' for $(n, m') = (n, m - 1)$ with the same length profile except that a_i and a_j are erased and $a'_k = \sqrt{a_i^2 + a_j^2}$ is inserted.¹³ To get from W' to a matrix W for the original profile $a_1, \dots, a_m$, erase the vector W'_k for a'_k in W' , and insert $W_i = \frac{a_i}{a'_k} W'_k$ and $W_j = \frac{a_j}{a'_k} W'_k$. One can check that the norms of W_i and W_j are respectively a_i and a_j and that this preserves the dot product of each pair of columns, hence preserving orthonormality. So W is a matrix with orthonormal columns and with desired row norms $a_1, \dots, a_m$.

Either way, we have shown that there is a matrix W of the desired form for (n, m) and any $a_1, \dots, a_m$ (with $a_i \leq 1$, and $\sum_{i=1}^m a_i^2 = n$) as well, completing the proof of the inductive step, and hence the proof. □

⁹The proof in words: if some W_i is too long compared to the other vectors in the configuration, then the direction W_i contains too much norm for all directions to contain the same norm (which is what it means for the configuration to be tight).

¹⁰We are using $[W'|W]$ to denote a block matrix with the $m \times (n - m)$ block W' and then $m \times n$ block W stacked horizontally.

¹¹The tight frame W constructed this way is called the *complement* of the tight frame W' . Or really, it is *a* complement, not *the* complement, because there is (in general) some freedom in how to complete an orthonormal set into an orthonormal basis. But the complement of a tight frame is unique up to isometry. (Or well, the usual definition also lets one rescale a complement freely. But it's unique up to isometry and rescaling.)

¹²One can see this, for instance, by noting that for a random pair of indices, $\mathbb{E}[a_i^2 + a_j^2] = \mathbb{E}[a_i^2] + \mathbb{E}[a_j^2] = 2 \frac{n}{m} \leq 1$.

¹³For convenience, we think of the other indices being unchanged by this, so the index set for W' is just missing i and j and it has an extra special index k .

2.4 A tight frame as a solution to a TMS-like optimization problem

As (almost) noted in [Elh+22], in the sparse limit, with biases set to zero, with the constraint that each $\|W_i\| = a_i$ for some particular constants $a_1, \dots, a_m \geq 0$, and with uniform importances, the TMS optimization problem is equivalent to minimizing $\sum_{i \neq j} \text{ReLU}(W_i \cdot W_j)^2$. A nicer-looking cousin of this is minimizing the *tight frame potential* $\sum_{i \neq j} (W_i \cdot W_j)^2$ with the same constraint $\|W_i\| = a_i$ for each $i \in [m]$. In addition to looking nice, its optima also turn out to have a simple characterization: they turn out to be tight frames.

Theorem 2.3 (The (generalized) Welch bound [Wel74][Wal18]).

$$\sum_{(i,j) \in [m] \times [m]} (W_i \cdot W_j)^2 \geq \frac{1}{n} \left(\sum_i \|W_i\|^2 \right)^2,$$

with equality if and only if W is a tight frame.

Proof. Consider the singular value decomposition $W = \sum_{i=1}^n \sigma_i u_i v_i^T$. Note that $WW^T = \sum_{i=1}^n \sigma_i^2 u_i u_i^T$. Let $\|A\|_F$ denote the Frobenius norm of A , i.e., $\|A\|_F = \sqrt{\sum_{i,j} A_{ij}^2}$. Using that the Frobenius norm is also equal to the sum of squared singular values of the matrix,

$$\sum_{(i,j) \in [m] \times [m]} (W_i \cdot W_j)^2 = \|WW^T\|_F^2 = \sum_{i=1}^n \sigma_i^4 \geq \frac{1}{n} \left(\sum_{i=1}^n \sigma_i^2 \right)^2 = \frac{1}{n} \|W\|_F^2 = \frac{1}{n} \left(\sum_i \|W_i\|^2 \right)^2,$$

where the inequality step is QM $\geq$ AM. Equality occurs iff all σ_i are equal, i.e. iff W is a tight frame. $\square$

In particular, with the constraints that $\|W_i\| = a_i$, Theorem 2.3 gives

$$\sum_{1 \leq i \neq j \leq m} (W_i \cdot W_j)^2 \geq \frac{1}{n} \left(\sum_i a_i^2 \right)^2 - \sum_i a_i^4,$$

still with equality iff W is a tight frame.

Corollary 2.3.1 (of Theorem 2.3 and Theorem 2.2). *For any sequence $a_1, \dots, a_m \geq 0$ with all $a_i^2 \leq \frac{1}{n} \sum_{j=1}^m a_j^2$, the global optima of*

$$\min_{\substack{W=(W_1, \dots, W_m) \in (\mathbb{R}^n)^m \\ \text{such that } \|W_1\|=a_1, \dots, \|W_m\|=a_m}} \sum_{1 \leq i \neq j \leq m} (W_i \cdot W_j)^2$$

are precisely the tight frames W .

Proof. By Theorem 2.2, there is a tight frame W with each $\|W_i\| = a_i$. Thus, the global minimum of $\sum_{1 \leq i \neq j \leq m} (W_i \cdot W_j)^2$ (subject to each $\|W_i\| = a_i$) is indeed $\frac{1}{n} \left(\sum_i a_i^2 \right)^2 - \sum_i a_i^4$, achieved iff W is a tight frame (with each $\|W_i\| = a_i$). $\square$

We now briefly discuss four extensions of the above, with connections to feature geometry.

Unbalanced lengths When the vectors W_i are constrained to have lengths a_i which make it impossible for them to form a tight frame (Theorem 2.2), i.e., if some $a_i^2 > \frac{1}{n} \sum_{j=1}^m a_j^2$, then the global optima of course cannot be tight frames. It turns out (Theorem 1 of [Cas+06]) that in this case, it is optimal to give vectors W_i with the largest lengths a_i their own orthogonal directions, continuing until the remaining squared lengths a_i are such that the inequality $a_i^2 \leq \frac{1}{n'} \sum_{\text{remaining } j} a_j^2$ is first satisfied for remaining indices i , with n' being the dimension of the orthogonal complement of the span of the k previously placed vectors (i.e., $n' = n - k$), and then placing the remaining vectors as a tight frame in this orthogonal complement.

Local optima In fact, by Theorem 1 of [Cas+06], any local minimum has the form specified in the previous paragraph. Since compactness of the optimization domain (which is $\cong (S^{n-1})^m$) implies that the tight frame potential has at least one (constrained) global minimum, and a global minimum must also be a local minimum, there is a (constrained) global minimum which has this form. One can check that any two configurations of this form have the same $\sum_{1 \leq i \neq j \leq m} (W_i \cdot W_j)^2$. It follows that all (constrained) local minima are also (constrained) global minima.¹⁴ The core of the argument from [Cas+06] characterizing these local optima is an analysis of the second-order behavior of the loss around its stationary points.¹⁵ We provide a variation of this argument in Appendix A.

One might hope that an analogous argument would also let one characterize the local minima of the restricted [Elh+22] loss, $\sum_{i \neq j} \text{ReLU}(W_i \cdot W_j)^2$; we now briefly explain an obstacle to such an argument. The proof of Theorem 1 of [Cas+06] begins by writing down the Lagrange multiplier equations for the tight frame potential, which are $\lambda_i W_i = \sum_{j \in [m]} (W_i \cdot W_j) W_j$ — this very helpfully implies that W_i are eigenvectors of the symmetric matrix $W^T W$ (details in Appendix A). However, for $\sum_{i \neq j} \text{ReLU}(W_i \cdot W_j)^2$, the Lagrange multiplier equations are instead $\lambda_i W_i = \sum_{i \neq j} \text{ReLU}(W_i \cdot W_j) W_j$

The weighted case As in [Elh+22], one can consider variants of this optimization problem where how much orthogonality matters can be different for different (pairs of) vectors. Assigning coefficients $I_1, \dots, I_m \in \mathbb{R}^{\geq 0}$, which we will call *importances*, respectively to the vectors $W_1, \dots, W_m$, a mathematically neat variant is to let the loss be $\sum_{1 \leq i \neq j \leq m} \sqrt{I_i I_j} (W_i \cdot W_j)^2$.¹⁶ This optimization problem can be reduced to the unweighted case by setting $W'_i = I_i^{1/4} W_i$ (and $a'_i = \sqrt{I_i} a_i$) — the loss then becomes $\sum_{1 \leq i \neq j \leq m} (W'_i \cdot W'_j)^2$. To get something more like the one in [Elh+22], we could also let the loss be $\sum_{1 \leq i \neq j \leq m} I_i (W_i \cdot W_j)^2$

Optimally nearly orthogonal vectors For unit vectors W_i , this gives $\sum_{1 \leq i \neq j \leq m} (W_i \cdot W_j)^2 \geq m \left(\frac{m}{n} - 1 \right)$. One can use this to get an upper bound on the minimal angle between two lines for a set of m lines in $\mathbb{R}^n$ [Wel74][SH03]:

$$\max_{(i,j), i \neq j} (W_i \cdot W_j)^2 \geq \frac{1}{m(m-1)} \sum_{1 \leq i \neq j \leq m} (W_i \cdot W_j)^2 \geq \frac{m \left(\frac{m}{n} - 1 \right)}{m(m-1)} = \frac{1}{n} \left(1 - \frac{n-1}{m-1} \right).$$

So there is a pair $i \neq j$ with $|W_i \cdot W_j| \geq \sqrt{\frac{1}{n} \left(1 - \frac{n-1}{m-1} \right)}$. In fact, there are a bunch of cases this simple bound is tight. For this to happen, both inequalities above must be equalities: the numbers $|W_i \cdot W_j|$ for all pairs $i \neq j$ must be equal, and W must be a tight frame.¹⁷

say what happens when not

3 Upper bounds on the sum of feature dims

Having introduced all the main concepts, we will now show that feature dimensionalities have a few intuitively reasonable properties, beginning from upper bounds on the sum of feature dimensionalities.

¹⁴The backward implication in Theorem 2.2 — i.e., that given that all $a_i^2 \leq \frac{1}{n} \sum_{j=1}^m a_j^2$, there is a tight frame with $\|W_i\| = a_i$ — is also a corollary [Cas+06].

¹⁵There are in fact more stationary points than just those that have the form specified above — for example (for certain length profiles), two nontrivial tight frames in orthogonal complements — but there turns out to be a neat way to slide out of any of the other ones.

¹⁶Of course, instead of $\sqrt{I_i I_j}$, one could have chosen the factor here to be $I_i I_j$ or $I_i^2 I_j^2$ instead. These are all the same up to redefining I_i — we went with $\sqrt{I_i I_j}$ to cohere with what comes later in this paragraph (and the analogous decision for what comes later is made so as to cohere with [Elh+22]).

¹⁷These are called *equiangular tight frames*.

3.1 A simple bound: fewer dims than features

For starters, it would be weird for a vector to ‘get more than one dimension’. And indeed,

$$\dim_i(W_1, \dots, W_m) = \frac{(W_i \cdot W_i)^2}{\sum_j (W_i \cdot W_j)^2} \leq \frac{(W_i \cdot W_i)^2}{(W_i \cdot W_i)^2} = 1.$$

It follows that ‘you can’t have m features cover more than m dimensions’:

$$\sum_{i=1}^m \dim_i W \leq m,$$

with equality iff the features are orthogonal.

3.2 A better bound: fewer dims than the ambient space

However, typically, we are working with $m > n$, in which case there is the following much better bound, corresponding to the fact we’d intuitively expect from dimensionalities that their sum is at most n :

Proposition 3.0.1 ([Sch+23]). *For a configuration $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$,*

$$\sum_{i=1}^m \dim_i W \leq n.$$

This is alluded to in [Elh+22] and stated and proved in [Sch+23]. Taking the SVD to be an elementary operation, the proof we give here is simpler than the proof in Appendix F of [Sch+23]. The proof is inspired by the connection between feature dimensionalities and leverage scores provided by [Zha22] — in fact, it seems essentially certain to us that the same argument is present in the literature on leverage scores,¹⁸¹⁹ but we have not looked into this literature in detail and in particular have (re)constructed the argument ourselves.

Proof. Consider W as a matrix with rows $W_1, \dots, W_m$ as earlier, with a singular value decomposition $W = U\Sigma V^T$. Let $U'_i \in \mathbb{R}^m$ denote row i of U with the last $m - \text{rank } W$ entries replaced by zeros, taken as a vector.²⁰ The SVD gives $W_i^T = (U'_i)^T \Sigma V^T$. Plugging this into the equation for feature dimensionality gives

$$\begin{aligned} \dim_i W &= \frac{(W_i \cdot W_i)^2}{\sum_j (W_i \cdot W_j)^2} = \frac{(W_i^T W_i)^2}{\|W W_i\|^2} = \frac{\left((U'_i)^T \Sigma V^T W_i\right)^2}{\|U \Sigma V^T W_i\|^2} = \left(U'_i \cdot \frac{\Sigma V^T W_i}{\|\Sigma V^T W_i\|}\right)^2 \\ &\leq \|U'_i\|^2 \left\| \frac{\Sigma V^T W_i}{\|\Sigma V^T W_i\|} \right\|^2 = \|U'_i\|^2. \end{aligned}$$

It follows that $\sum_{i=1}^m \dim_i W \leq \sum_{i=1}^m \|U'_i\|^2 = \sum_{i=1}^{\text{rank } W} \|u_i\|^2 = \text{rank } W \leq n$.²¹ □

3.3 Efficient packings

We will now turn to examining vector configurations for which the previous bound is achieved.

Definition 3.1. We say a vector configuration $W = (W_1, \dots, W_m) \in (\mathbb{R}^n)^m$ is *efficient* if $\sum_{i=1}^m \dim_i W = n$.

¹⁸In fact, the term $\|U'_i\|^2$ that shows up in the proof is precisely the leverage score of row i of the matrix W .

¹⁹That said, it seems likely that the proof of Theorem 3.1 we provide later is not present in this literature.

²⁰Just like earlier, even though U'_i is a row of U , when we treat it as a matrix, we take it to be a column vector.

²¹The connection of this argument to leverage scores is the following. The leverage score of row i of the matrix W is defined to be $\max_{x \in S^{n-1} \subseteq \mathbb{R}^n} \frac{(Wx)_i^2}{\|Wx\|^2} = \max_{x \in S^{n-1} \subseteq \mathbb{R}^n} \left(U'_i \cdot \frac{\Sigma V^T x}{\|\Sigma V^T x\|} \right)^2$. Since U'_i is in the range of the linear map ΣV^T , the leverage score of row i is just $\|U'_i\|^2$, so the sum of leverage scores of the rows of W is $\text{rank } W$.

[Sch+23] prove that optima of the feature packing problem they study are efficient, and TMS configurations are also often efficient (TODO: is this true?). Our proof of $\sum_{i=1}^m \dim_i W = n$ extends to the following characterization of efficient configurations:

Theorem 3.1 ([Sch+23]). *A configuration W is efficient $\iff$ it spans $\mathbb{R}^n$ and there are orthogonal subspaces $V_1, \dots, V_\ell$ of $\mathbb{R}^n$ such that:²²*

- For each $i \in [m]$, there is a $k \in [\ell]$ such that $W_i \in V_k$.
- For each $k \in [\ell]$, the subconfiguration in V_k is a tight frame in V_k .

Proof of $\implies$. From the proof of Proposition 3.0.1, $\sum_{i=1}^m \dim_i W = n$ implies that $\text{rank } W = n$, so W spans $\mathbb{R}^n$. To prove the rest, we will start by defining the orthogonal subspaces in which the subconfigurations will turn out to live. Coming back to the singular value decomposition $W = \sum_{i=1}^n \sigma_i u_i v_i^T$, let $a_1, \dots, a_{\ell-1}$ be the indices after which there are strict decreases in the sequence $\sigma_1, \dots, \sigma_n$, i.e.,

$$\sigma_1 = \sigma_2 = \dots = \sigma_{a_1} > \sigma_{a_1+1} = \dots = \sigma_{a_2} > \dots > \sigma_{a_{\ell-1}} = \dots = \sigma_n$$

and, for convenience, let $a_0 = 0$ and $a_\ell = n$. Let $V_1, \dots, V_\ell$ be spans of the corresponding right singular vectors, i.e., for each $1 \leq k \leq \ell$, we let

$$V_k := \text{span} \{v_{a_{k-1}+1}, v_{a_{k-1}+2}, \dots, v_{a_k}\}.$$

Let us now argue that each W_i is in one $V_1, \dots, V_\ell$. From the proof of Proposition 3.0.1, $\sum_{i=1}^m \dim_i W = n$ implies that for each i , there is $\lambda_i \in \mathbb{R}$ such that $U'_i = \lambda_i \Sigma V^T W_i$. Plugging in $W_i = V \Sigma^T U'_i$, this becomes

$$U'_i = \lambda_i \Sigma \Sigma^T U'_i.$$

Let $I_i \subseteq \{1, 2, \dots, n\}$ denote the set of indices at which U'_i has nonzero entries. Note that $\lambda_i \Sigma \Sigma^T$ is an $m \times m$ diagonal matrix: the first entries along the diagonal are (in order) $\lambda_i \sigma_1^2, \dots, \lambda_i \sigma_n^2$, and all other entries are 0. For the equation above to hold, the diagonal entries at indices in I_i need to be equal. Since $W_i \neq 0$, so $U'_i \neq 0$, so $\lambda_i \neq 0$, it follows that the diagonal entries of $\Sigma \Sigma^T$ at indices in I_i are also all equal, so they all have some particular value σ^2 . Thus, there is a $1 \leq k \leq \ell$ such that $I_i \subseteq \{a_{k-1} + 1, a_{k-1} + 2, \dots, a_k\}$ (with a_k as defined earlier). It follows that $W_i = V \Sigma^T U'_i \in V_k$.

We've established that each W_i is in one of the orthogonal subspaces V_k . Note that any two different V_k are spanned by disjoint sets of right singular vectors, hence orthogonal to each other. Furthermore, the subconfiguration in each V_k is a tight frame. This is because, for a particular V_k , letting W' denote the submatrix of W where only the rows with $W_i \in V_k$ are kept, for any unit vector $v \in V_k$, we have $\|W'v\|^2 = \|Wv\|^2 = v^T V \Sigma^T \Sigma V^T v = \sigma_{a_k}^2$ which by Lemma 2.1 implies that the subconfiguration is a tight frame. $\square$

Proof of $\impliedby$. $\square$

In fact, a moderate amount of additional effort extends the above proof to the proof of a corresponding stability²³ theorem:

Theorem 3.2.

We give a proof in Appendix B.

4 Symmetric configurations

Definition 4.1. Let us say that a configuration is isogonal https://en.wikipedia.org/wiki/Isogonal_figure

²²In Anthropic's terminology, instead of the conditions below, we'd say W is a tegum product of tight frames.

²³We are using stability here in the sense of, for instance, [KRS21].

isogonality implies equal feature dims
given isogonality, efficient $\Rightarrow$ isotropic
stronger symmetry condition implies isotropic
can do root systems by restricting to roots of a certain length. weyl group transitive to begin with so still transitive. also still root system, still can reflect across. also need basis though?
weyl group transitive roots of a particular length form another root system they also span the space because weyl group acts transitively $\Rightarrow$ for any subspace, there is a weyl group element which does not fix that subspace

5 Open problems

Here are some problems that seem interesting/promising to us:

1. Provide a realistic and reasonably wide class of feature packing problems for which optimal configurations are:
 - (a) tight frames (Definition 2.1);
 - (b) efficient (Definition 3.1).

That is, together with a specification of this class, provide either empirical or conceptual evidence — of course, preferably, proofs — of its optima indeed being tight frames / efficient packings. Alternatively, provide empirical/conceptual evidence that optimal feature configurations in toy models, or feature configurations in real models, have nothing to do with tight frames or efficient packings.

2. Come up with a more principled heuristic model of the ‘potential’/‘loss’/‘forces’, preferably one that’s plausibly robust to details of the architecture/task/training data, that control what feature configurations show up in real models. Study the configurations that are optimal in this setting mathematically or empirically. (For example, the tight frame potential satisfies the second criterion: we have a good understanding of its minimizers, namely, it turns out that all local minimizers — configurations in equilibrium under the force (called the *frame force*) corresponding to the frame potential — are always tight frames (see [Cas+06]), which are always global minimizers (as we saw). What’s more, we have a good understanding of the set of all tight frames (see [Cah+11]). But we don’t really have any good conceptual story for expecting feature configurations to minimize the tight frame potential. The situation is similar for feature dimensionalities.)
3. Propose feature decomposition methods²⁴ that work conditional on activations being sparse linear combinations of features from a tight frame / efficient packing which leverage these properties (that is, work better than unconditional methods conditional on these properties being (nearly) satisfied).

6 Acknowledgments

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²⁴See e.g. [HMK23] for an explanation of what we mean by feature decomposition methods.

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A Local optima of the tight frame potential[Cas+06]

In this section, we give a variation of the characterization of all local optima of the tight frame potential from [Cas+06]. Informally, the characterization says that for any local minimum, the longest W_i are placed orthogonally until the remaining vectors can be placed as a tight frame in their orthogonal complement.²⁵

Theorem A.1 ([Cas+06]). *Fix (a dimension) $n \in \mathbb{N}$, and (lengths) $a_1 \geq a_2 \geq \dots \geq a_m \geq 0$,²⁶ and let $k \geq 0$ be the smallest integer such that $a_{k+1}^2 \leq \frac{1}{n-k-1} \sum_{i=k+2}^m a_i^2$.²⁷ Then, any (constrained) local minimum of $\sum_{1 \leq i < j \leq m} (W_i \cdot W_j)^2$ with $W_1, \dots, W_m \in \mathbb{R}^n$ subject to the constraints $\|W_1\| = a_1, \dots, \|W_m\| = a_m$ satisfies the following:*

- $W_1, W_2, \dots, W_k$ are orthogonal.
- $W_{k+1}, \dots, W_m$ form a tight frame in the orthogonal complement of $\text{span}\{W_1, \dots, W_k\}$.

Proof from [Cas+06]. First, if we can prove the result with all $a_i > 0$, then the case with some additional $a_j = 0$ introduced follows — local minima of the two optimization problems are in bijection, and we can always think of the $W_i = 0$ as being in the tight frame. So we’ll assume all $a_i > 0$ from now on. Second, notice that at a local minimum, the dimension of the span of $W_1, \dots, W_m$ is $\min(m, n)$ — otherwise we could pick a vector W_i with nonzero dot product with at least one other W_j and slide it slightly into an

²⁵Indeed, this restatement is formally equivalent to the theorem given Theorem 2.2, but we want to refrain from stating it this way formally, because we would like it to be explicit that the theorem below really provides an independent proof of the backward implication of Theorem 2.2.

²⁶We’ll be assuming WLOG that the vectors are ordered in decreasing order of length (this just makes the statement cleaner).

²⁷Note that such an index exists, since $k = m - 1$ certainly works.

unoccupied orthogonal direction $h \in \mathbb{R}^n$, i.e., replacing W_i with $W_i(\varepsilon) = \sqrt{1 - \varepsilon^2}W_i + \varepsilon a_i h$ — it's clear that this locally decreases all nonzero $W_i \cdot W_j$ (with this i and $j \neq i$) (and leaves everything else unchanged),²⁸ and hence locally decreases $\sum_{i \neq j} (W_i \cdot W_j)^2$. This already implies that when $m \leq n$, any local optimum is a tuple of orthogonal vectors, completing the proof in that case. It remains to consider the case with $m > n$ and all a_i nonzero, in which case we've established that $W_1, \dots, W_m$ span $\mathbb{R}^n$.

Then²⁹, from the Lagrange multiplier theorem (skipping the calculation), one gets that any (constrained) local optimum has to satisfy $\lambda_i W_i = \sum_{j \in [m]} (W_i \cdot W_j) W_j$ for each $i \in [m]$, which can also be written as $(W^T W) W_i = \lambda_i W_i$. In words, each W_i is an eigenvector of $W^T W$. Since $W^T W$ is positive-definite,³⁰ it has positive real eigenvalues and its eigenspaces E_λ for different eigenvalues are orthogonal. Using orthogonality, we can decompose $(W^T W) W_i = \sum_{j \in [m] \text{ such that } \lambda_j = \lambda_i} (W_j W_j^T) W_i = \lambda_i W_i$, whence we see that $\sum_{j \in [m] \text{ such that } \lambda_j = \lambda_i} (W_j W_j^T)$ acts as multiplication by λ_i on the span of all W_i with eigenvalue λ_i . Hence, all W_i with this particular eigenvalue form a tight frame in their span, with frame bound λ_i . Since $W_1, \dots, W_m$ span $\mathbb{R}^n$, the span of all W_i with eigenvalue λ_i is the eigenspace E_{λ_i} . **It also follows that the vectors W_i are partitioned into the eigenspaces E_λ of the matrix $W^T W$, with the subconfiguration in each eigenspace being a tight frame in that eigenspace.**

Let the smallest eigenvalue be $\lambda > 0$, and let some other eigenvalue be $\lambda' > \lambda$. We now get to the cleverest idea in this argument: it turns out that if the W_i in $E_{\lambda'}$ are linearly dependent, then there is a way to jointly slide them towards E_λ while locally decreasing loss. We pick a unit vector $h \in E_\lambda$ to slide towards and speeds c_i , potentially different for different W_i , to be chosen later. That is, for all W_i with eigenvalue λ' , we simultaneously replace W_i with (as a function of ε) $W_i(\varepsilon) = \sqrt{1 - \varepsilon^2 c_i^2} W_i + \varepsilon c_i a_i h$.³¹ In the loss $\sum_{i,j} (W_i \cdot W_j)^2$, this changes only the following two kinds of terms:

1. Cross-terms, i.e. squared dot products of some $W_i(\varepsilon)$ against some vector $W_j \in E_\lambda$. The additional loss these contribute is

$$+2 \sum_{(i,j) \text{ with } W_i \in E_{\lambda'}, \text{ and } W_j \in E_\lambda} (W_i(\varepsilon) \cdot W_j)^2 = 2 \sum_{i \text{ with } W_i \in E_{\lambda'}} \varepsilon^2 c_i^2 a_i^2 \sum_{W_j \in E_\lambda} (h \cdot W_j)^2 = 2\varepsilon^2 \lambda \sum_{i \text{ with } W_i \in E_{\lambda'}} c_i^2 a_i^2,$$

where on the last step we used the fact that the vectors in E_λ form a tight frame with frame bound λ .

2. Terms between $W_i(\varepsilon)$ and $W_j(\varepsilon)$ now contribute

$$\begin{aligned} & \sum_{W_i, W_j \in E_{\lambda'}} \left(\left(\sqrt{1 - \varepsilon^2 c_i^2} W_i + \varepsilon c_i a_i h \right) \cdot \left(\sqrt{1 - \varepsilon^2 c_j^2} W_j + \varepsilon c_j a_j h \right) \right)^2 \\ &= \sum_{W_i, W_j \in E_{\lambda'}} \left(\sqrt{(1 - \varepsilon^2 c_i^2)(1 - \varepsilon^2 c_j^2)} W_i \cdot W_j + \varepsilon^2 (c_i a_i)(c_j a_j) \right)^2 \\ &= \sum_{W_i, W_j \in E_{\lambda'}} (W_i \cdot W_j)^2 - 2\varepsilon^2 \sum_{i \text{ with } W_i \in E_{\lambda'}} c_i^2 \sum_{W_j \in E_{\lambda'}} (W_i \cdot W_j)^2 + 2\varepsilon^2 \sum_{W_i, W_j \in E_{\lambda'}} (c_i a_i W_i) \cdot (c_j a_j W_j) \\ &= \sum_{W_i, W_j \in E_{\lambda'}} (W_i \cdot W_j)^2 - 2\varepsilon^2 \lambda' \sum_{i \text{ with } W_i \in E_{\lambda'}} c_i^2 a_i^2 + 2\varepsilon^2 \left(\sum_{W_i \in E_{\lambda'}} c_i a_i W_i \right)^2. \end{aligned}$$

We now use the fact that the collection of $W_i \in E_{\lambda'}$ is linearly dependent to choose the coefficients c_i such that $\sum_{W_i \in E_{\lambda'}} c_i a_i W_i = 0$ such that not all c_i are 0. This makes the last term above 0, so the change in loss from cross terms is just $-2\varepsilon^2 \lambda' \sum_{i \text{ with } W_i \in E_{\lambda'}} c_i^2 a_i^2$.

Putting these terms together, we see that sliding all $W_i \in E_{\lambda'}$ towards E_λ together in this way changes loss by $-2\varepsilon^2 (\lambda' - \lambda) \sum_{i \text{ with } W_i \in E_{\lambda'}} c_i^2 a_i^2 < 0$ (where we used the fact that $\lambda' > \lambda$). Hence, the W_i in $E_{\lambda'}$ must be linearly independent. Since we already knew they also form a tight frame with frame bound λ' , the first

²⁸There is no effect up to order ε , but there is a negative contribution of order ε^2 .

²⁹well, after checking that the jacobian of the constraint function has full rank at any point in the domain

³⁰see https://en.wikipedia.org/wiki/Definite_matrix#Decomposition

³¹Note that $\|W_i(\varepsilon)\| = a_i$.

paragraph of Section 2.2 then tells us that **the $W_i \in E_{\lambda'}$ must be the rescaling of an orthonormal basis for $E_{\lambda'}$ by $\sqrt{\lambda'}$.**

At this point, we have already established that any local optimum must have a bunch of vectors placed orthogonally — let's say that the number of these is k' — and the rest placed as a tight frame in their orthogonal complement. To finish the proof of the theorem, it suffices to show that the vectors placed orthogonally must be precisely $W_1, \dots, W_k$, where $k \geq 0$ be the smallest integer such that $a_{k+1}^2 \leq \frac{1}{n-k-1} \sum_{i=k+2}^m a_i^2$.

The following **auxiliary claim** will be helpful: **for all indices $k' \geq k+1$, we have $a_{k'}^2 \leq \frac{1}{n-k'} \sum_{i=k'+1}^m a_i^2$.** The claim follows from $(n-k')a_{k'}^2 \leq (n-k-1)a_{k+1}^2 - \sum_{i=k+2}^{k'} a_i^2 \leq \sum_{i=k'+1}^m a_i^2$.

Now, consider the largest index $1 \leq j \leq m$ with $W_j \notin E_{\lambda}$ (if there is no such index j , then everything is in the tight frame, so we are already done). With $W_j \in E_{\lambda'}$, we saw in the previous paragraph that $a_j = \sqrt{\lambda'}$, and thus

$$a_j^2 = \lambda' > \lambda = \frac{1}{n-k'} \sum_{i \text{ with } W_i \in E_{\lambda}} a_i^2 = \frac{1}{n-k'} \left(\sum_{i < j \text{ with } W_i \in E_{\lambda}} a_i^2 + \sum_{i=j+1}^m a_i^2 \right).$$

Using the fact that the sequence a_i is non-increasing to lower-bound the first sum, it follows that

$$(n-k')a_j^2 > (j-k')a_j^2 + \sum_{i=j+1}^m a_i^2, \quad \text{so} \quad (n-j)a_j^2 > \sum_{i=j+1}^m a_i^2.$$

From the auxiliary claim above, we conclude that $j \leq k$. **Thus, $W_{k+1}, W_{k+2}, \dots, W_m \in E_{\lambda}$.**

It remains to show that there is no $j \leq k$ with $W_j \in E_{\lambda}$. Supposing for a contradiction that there is one, let $j \leq k$ be the largest such index, and let j' be the smallest. Then $(n-k)a_j^2 \geq (n-k)a_{j'}^2 > \sum_{i=k+1}^m a_i^2$. Using the auxiliary claim on the subsequence of those a_i with $W_i \in E_{\lambda}$ lets us conclude from this also that $a_{j'}^2 > \frac{1}{\dim(E_{\lambda})} \sum_{i \text{ with } W_i \in E_{\lambda}} a_i^2$. By the forward implication of Theorem 2.2, this contradicts with the sub-configuration in E_{λ} being a tight frame. Thus, the vectors in E_{λ} are indeed precisely $W_{k+1}, W_{k+2}, \dots, W_m$, and these form a tight frame in E_{λ} ; and $W_1, \dots, W_k$ are placed orthogonally. $\square$

B Stability of Theorem 3.1

Here's the version of the